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Psychometric Properties of the Romanian Version of Experiences in Close Relationships-Revised Questionnaire (ECR-R)

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Abstract

The present research investigated the psychometric properties of the Romanian version of the Experience in Close Relationships-Revised questionnaire (ECR-R; Fraley, Waller & Brennan, 2000). Based on exploratory factor analysis, the first study ($N = 245$) replicated a dual-factor solution. The same factorial structure was sustained by the fit indices obtained in the confirmatory factor analysis (CFA) conducted in the second study ($N = 220$). After excluding five items – due to inadequate factor loadings - a 31 item Romanian version of the ECR-R scale resulted with 14 items measuring avoidance attachment and 17 items measuring anxiety attachment. Both scales are highly reliable and convergent with RQ (Bartholomew and Horowitz, 1991) measures. Further validation studies are needed.

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1. Introduction

Conceptualized as a lifespan personality theory, attachment theory describes the way in which initial social interactions lay the base for the internal working models of self in relationship with the caregiver (Bowlby, 1973). These models allow the person to anticipate the reactions of others across life situations. Due to expectancy-behavior-reaction-confirmation circularity, the internal working models are strengthened and become a stable and integrating part of the adult's life (Scharfe & Cole, 2006; Shorey, 2010). The interaction of two distinct dimensions: the Model of the Self and the Model of the Other, are theorized by Bartholomew (1990). This interaction defines four different attachment styles: secured, preoccupied, avoidant and fearful. The two

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dimensions dealing with general expectations about own value and other's availability while attachment styles are meta-strategies of dyadic emotion regulation (Sroufe, 1996; Griffin & Bartholomew, 1994b).

Two concurrent directions dispute their efficiency in adult attachment measurement. The dimensional approach (anxiety and avoidance) implies that people can be quantitatively ordered. Any new quality resulting from the interaction between the two variables is lost. However, this approach is advantageous as it doesn't lose much information like when separating people in distinct groups. Dimensions allow access to regressions and structural equations (Griffin & Bartholomew, 1994b; Shorey, 2010). The discrete categories approach (attachment styles) implies that each individual can be placed in the most appropriate group, directly or by the means of dimensional scores. This approach is better suited to identify the unique strategies of each attachment style. However, it implies that people belong naturally to discrete categories and that only intergroup variance is meaningful. This overemphasizes in-group similarity and inter-group difference to the point of self-confirmatory research (Griffin & Bartholomew, 1994b).

The Experiences in Close Relationship Scale (ECR) was created by Clark and Shaver (1998) and has been adapted in Romania by Negrei and Sava (2006). However, through an analysis based on item-response theory, Fraley, Waller and Brennan (2000) generated a revised scale (ECR-R) with even better psychometric properties, which remained among the best measures of Adult Attachment (Shorey, 2010).

The present research seeks to adapt the ECR-R questionnaire in the Romanian cultural context. Both studies of the present research are examining the psychometric properties of the Romanian version of the ECR-R.

2. Study 1

2.1. Methodology

2.1.1. Participants

The study was conducted on a volunteer convenience sample. The total number of participants ($N = 245$) consisted of students enrolled in the Faculty of Psychology and Educational Sciences. They were 211 females (86.1%) and 33 males (13.5%) with a mean age of 25.07 ($SD = 6.75$).

2.1.2. Instruments and procedure

Participants completed the Experiences in Close Relationships-Revised (ECR-R; Fraley et al., 2000) and Relationship Questionnaire (RQ; Bartholomew & Horowitz, 1991).

Fraley et al.'s (2000) ECR-R contains 18 items assessing romantic attachment anxiety and 18 items assessing romantic attachment avoidance. The translation procedure was conducted based on back-translation and using the expressions "iubit(ă)" (lover) for the original "romantic partner" and "prieten(ă)" (friend) for the original "partner". We also found a better equivalent for the expression "to depend on" – "a se bizui pe" (to rely on). We chose a six points scale in order to avoid the central tendency. Internal consistency alpha levels ranged between .89 and .91.

Bartholomew and Horowitz's (1991) RQ contains four assertions measuring four attachment prototypes (secure, dismissing, preoccupied, and fearful). We computed ratings of attachment avoidance and anxiety using the following formulae (Griffin and Bartholomew, 1994b): RQ model of others = (dismissing + fearful prototypes) – (secure + preoccupied prototypes); RQ model of self = (preoccupied + fearful prototypes) – (secure + dismissing prototypes).

2.2. Results and discussions

ECR-R scales' results are normally distributed and the internal consistency alphas are showing excellent reliability for both scales (Avoidance: $M = 2.55$, $SD = 0.84$, $Skewness = .24$, $Kurtosis = -.63$, Cronbach's $\alpha = .89$;

Anxiety: $M = 2.37$, $SD = 0.94$, $Skewness = .80$, $Kurtosis = .27$, Cronbach's $\alpha = .91$). Anxiety and avoidance showed a moderate and positive correlation ($r = .35$, $p < .01$).

Table 1. Factor Loadings for the ECR-R and RQ Measures

Measure	F1	F2
<i>Avoidance</i>		
I prefer not to show a partner how I feel deep down. / Prefer să nu arăt prietenilor ce simt eu cu adevărat.		.646
I feel comfortable depending on romantic partners. / Mi-e ușor să mă bizui pe iubii/iubitele mele. (r)		.476
I prefer not to be too close to romantic partners. / Prefer să nu fiu prea apropiat(ă) de iubii/iubitele mele.		.491
I find it relatively easy to get close to my partner. / Mi se pare relativ ușor să mă apropii de prieteni. (r)		.536
It helps to turn to my romantic partner in times of need. / Mi-e de ajutor să mă adresez, la nevoie, iubitului/iubitei mele. (r)		.533
I find it easy to depend on romantic partners. / Mă simt în largul meu să mă bizui pe iubii/iubitele mele. (r)		.602
I tell my partner just about everything. / Spun prietenilor aproape tot. (r)		.741
It's not difficult for me to get close to my partner. / Nu mi-e greu să mă apropii de prieteni. (r)		.561
I talk things over with my partner. / Discut tot felul de lucruri cu prietenii. (r)		.650
I am nervous when partners get too close to me. / Mă simt neplăcut când prietenii se apropie prea mult de mine.		.659
It's easy for me to be affectionate with my partner. / Mi-e ușor să fiu afectuos / afectuoasă cu prietenii. (r)		.734
I usually discuss my problems and concerns with my partner. / De regulă, îmi discut problemele și grijiile cu prietenii. (r)		.752
I feel comfortable sharing my private thoughts and feelings with my partner. / Mă simt în largul meu să împărtășesc cu prietenii gândurile și emoțiile mele intime. (r)		.653
I am very comfortable being close to romantic partners. / Mi-e confortabil să mă apropii de iubii/iubitele mele. (r)		.533
RQ model of others		.663
<i>Anxiety</i>		
I'm afraid that I will lose my partner's love. / Mi-e teamă că voi pierde afecțiunea prietenilor mei.		.711
I often worry that my partner will not want to stay with me. / Îmi fac deseori griji că prietenii nu vor dori să stea cu mine.		.777
I often worry that my partner doesn't really love me. / Îmi fac deseori griji că prietenii nu tin cu adevărat la mine.		.729
I worry that romantic partners won't care about me as much as I care about them. / Îmi fac griji că iubii/iubitele mele nu vor ține la mine la fel de mult cât țin eu la ei/ele.		.727
Sometimes romantic partners change their feelings about me for no apparent reason. / Uneori, prietenii își schimbă sentimentele față de mine fără niciun motiv vizibil.		.697
I often wish that my partner's feelings for me were as strong as my feelings for him or her. / Îmi doresc deseori ca sentimentele prietenilor mei pentru mine să fie la fel de puternice ca sentimentele mele pentru ei.		.570
I worry a lot about my relationships. / Îmi fac multe griji în legătură cu relațiile mele cu oamenii.		.653
When I show my feelings for romantic partners, I'm afraid they will not feel the same about me. / Când arăt ce simt față de prieteni, mi-e teamă că ei nu vor simți la fel pentru mine.		.795
My romantic partner makes me doubt myself. / Iubitul meu / iubita mea mă face să am îndoieli legate de mine însumi/însămi.		.595
I do not often worry about being abandoned. / Nu-mi fac prea des griji că voi fi abandonat(ă). (r)		.510
I find that my partner(s) don't want to get as close as I would like. / Mi se pare că prietenii nu vor să se apropie atât cât mi-as dori.		.798
My desire to be very close sometimes scares people away. / Dorința mea de a fi foarte apropiat(ă) uneori îi sperie pe oameni.		.541
I'm afraid that once a romantic partner gets to know me, he or she won't like who I really am. / Mi-e teamă că, de îndată ce iubitul/iubita va ajunge să mă cunoască, nu-i va plăcea cum sunt eu de fapt.		.605
It makes me mad that I don't get the affection and support I need from my partner. / Mă înfurie faptul că nu primesc afecțiunea și sprijinul de care am nevoie de la prieteni.		.713
My partner only seems to notice me when I'm angry. / Prietenii par să mă observe doar când sunt furios/furioasă.		.533
When my partner is out of sight, I worry that he or she might become interested in someone else. / Când prietenii nu sunt cu mine, îmi fac griji că ar putea fi interesați de alte persoane.		.713
I worry that I won't measure up to other people. / Mi-e teamă că nu mă voi ridica la înălțimea așteptărilor prietenilor mei.		.803
RQ model of self		.665

(r) indicates reverse scored items. Item with less than .30 loadings are not reported. The first assertion represents the original item / the second is the Romanian version.

Further, we conducted a principal components exploratory factor analysis with varimax rotation on the ECR-R. Although five factors with eigenvalues greater than one were displayed, the steeply decreasing trend after the

second factor (9.82, 4.38, 2.11, 1.48, and 1.2) revealed a two-factor solution that explained 46% of the total variance. Due to very low saturation we decided to exclude item 6 from the scale and we also excluded items 7, 29, 32, and 36 because were highly saturated in both factors. After the modification, a 31 item Romanian version of the ECR-R scale resulted. The rotated factor loadings for ECR-R items can be seen in Table 1.

To verify the assertion that ECR-R and RQ scales are measuring the same construct, we ran another exploratory factor analysis with the ECR-R 31 items and the RQ computed models of avoidance and anxiety. RQ measure of avoidance loaded solely on the same factor with the ECR-R items of avoidance and EQ measure of anxiety loaded solely on the same factor with the ECR-R items of anxiety.

In line with previous research that explored the psychometric properties of the ECR-R (Sibley & Liu, 2004; Sibley et al., 2005) our data revealed a two-factor structure and high reliability indices.

3. Study 2

3.1. Methodology

3.1.1. Participants

In the second study we also used a student volunteer sample, recruited from the Faculty of Psychology and the Faculty of Law. The final sample consisted in 220 participants with a mean age of 22.30 ($SD = 1.25$), most of them being law students (77.7%) and females (76.4%).

3.1.2. Measures and procedures

In this study, participants completed the Experiences in Close Relationships-Revised (ECR-R; Fraley et al., 2000) 31 items questionnaire, on a six point scale. The romantic attachment avoidance 14 items scale's internal consistency alpha is .86 and the romantic attachment anxiety 17 items scale's alpha is .88, thus, the reliability coefficients are consistent with previous results. Participants also received the Relationship Questionnaire (RQ; Bartholomew & Horowitz, 1991).

3.2. Results and discussions

Confirmatory factor analysis (CFA) with item parcels was conducted to test the appropriateness of the two-factor model. Avoidance scale was aggregated in 5 item parcels and anxiety scale was parceled in 6 groups of items. The indices suggest an excellent fit of the two-factor solution: χ^2 (43, $n=220$) = 84.7, $p < .01$; GFI = .93; AGFI = .90; NFI = .93; CFI = .96; RMSEA = .06. All standardized path coefficients for the two-factor solution ranged between 0.70 and 0.90 for item parcels computed for avoidance and between 0.58 and 0.84 for item parcels assessing anxiety, and were statistically significant. The two latent factors (avoidance and anxiety) correlated at 0.20.

To verify the convergence between avoidance and anxiety factors measured with the ECR-R scales and the RQ self and others models, we introduced the RQ measures in the model. The results indicated an excellent fit: χ^2 (63, $n=220$) = 114.1, $p < .01$; GFI = .92; AGFI = .89; NFI = .92; CFI = .96; RMSEA = .06. The standardized path coefficient for the relation between RQ avoidance and ECR-R avoidance latent factor equals .57 ($p < .05$, $R^2 = .32$), and for RQ anxiety and ECR-R anxiety latent factor equals .55 ($p < .05$, $R^2 = .31$).

Using the same approach as Sibley and Liu (2004), and Sibley et al. (2005) - confirmatory factor analysis with item parcels – the dual-factor solution was once again sustained with very good fit indices.

4. General discussion

We assessed the factor structure, internal consistency and convergent validity of the ECR-R instrument. Consistent with previous research (Sibley & Liu, 2004; Sibley et al., 2005), the two factor solution representing dimensions of attachment anxiety and avoidance was the most appropriate. Separate exploratory and confirmatory factor analyses performed on the data collected in Study 1 and Study 2 sustained this conclusions. The internal consistency coefficients suggest high reliability for both scales, reaching values between .86 and .89 for avoidance scale, and between .88 and .91 for anxiety scale. These results also indicated that the ECR-R and RQ assessed the same pair of attachment dimensions (Griffin & Bartholomew, 1994b; Sibley et al., 2005).

The present work represents the first steps in the cultural adaptation of the ECR-R instrument in Romania. Future research could focus on temporal reliability and other forms of validity, and further explore the psychometric properties on different and more balanced samples. The findings that we reported in this paper provide important information on the psychometric properties of the questionnaire and suggest that it maintains acceptable classical psychometric properties.

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